A DISASTER WAITING TO HAPPEN

How the North Bengal floods unfolded and future steps that can stop such debacles

Graphic by
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SLIPPERY SLOPES

In hilly Darjeeling and Kalimpong, mountainsides are prone to landslides as they are made of shale and sandstone and heavy annual rainfall. It is worsened by urbanisation, construction

EFFECT

- When over 300 mm of rain fell on Oct. 5-6, it triggered landslides that tore into tea gardens, roads, settlements, and injected sediment into rivers, worsening floods downstream

DENUDED FORESTS

Forests in Himalayan foothills (Dooars) in Darjeeling, Jalpaiguri, Cooch Behar slowed rainwater runoff. They are now pierced by tea gardens, settlements

EFFECT

- During monsoon, water gushes through deforested areas and into rivers faster, carrying silt and debris that reduce channel capacity
- In early October, rain water hurtled down unchecked to the rivers, which could not contain it

WAYS TO SAVE THE SITUATION

- Human encroachment in abandoned river channels must be stopped; land use in floodplains strictly regulated
- Urban planning in Darjeeling, Kalimpong towns muct integrate landslide susceptibility maps that mark high-risk zones
- Roads, settlements must avoid blocking natural drainage; river banks need buffer zones
- ISRO's Persistent Scatterer InSAR, which detects slope movements, can provide early landslide warnings
- Remote sensing, machine learning can map land-use changes over time, guiding targeted interventions